High-Frequency Market Making Infrastructure: Part One on the Data Caliber of Six Types of Market Quotes

SK (@StepOneAi) · 7 August 2026 · posted on X · translated from Chinese

When we were working on stuff related to funding rates, the earliest pit we fell into was very straightforward: we pulled the funding rates of different contracts onto one table for side-by-side comparison, only to find the conclusions unstable. For the same stretch of market action, with similar deviations, some contracts' rates would spike high, while others barely budged. Our instinctive explanation at the time was that sentiment differed.

It was only later, after reading the docs, that we realized the issue wasn't sentiment—it was that we weren't comparing the same thing. Funding rates aren't natural market readings; they're a measurement taken by the platform according to a set of rules. Those rules include a few configurable, adjustable components, but we were defaulting to treating them as constants.

This piece breaks down that set of rules. For all the specific terms mentioned below, please refer to the contract documentation for whatever you're actually trading.

First off, the rate structure for mainstream perpetuals is broadly consistent: `F = average of the premium index + clamp(interest rate - premium index, upper and lower bounds)`. The differences hide in the averaging.

The common approach is to sample the premium index at fixed intervals, then do a time-weighted average over the entire settlement period. The number of points involved in the average is determined jointly by the sampling interval and the settlement period: `n = settlement period length / sampling interval`.

Here's the first thing worth noting: the settlement period is a variable, not the default 8 hours. Plenty of platforms set the period per contract and will temporarily adjust it under specific conditions; some APIs even have a dedicated settlement period hours field, and its very existence signals that it can change. But funding rate history APIs usually only return the rate and timestamp, not the period. If you align historical sequences assuming a fixed 8 hours, you'll get thrown off the moment you cross a period adjustment.

The second thing is even easier to overlook: even for the same time-weighted average, the weight functions aren't necessarily identical.

We've seen at least two kinds. One type of doc just mentions "time-weighted average" and stops there, without giving an explicit weight formula. The other spells out the formula directly, with weights that increase linearly: $(P_1 \times 1 + P_2 \times 2 + ... + P_n \times n) / (1+2+...+n)$. Under the latter, the weight of that final sampling point is n times the first one's. For an 8-hour period with one point per minute, n is 480.

The behavioral difference between these two is huge. Any judgment involving "how much impact a deviation right before settlement will have on the final rate" can't be carried over across platforms: in places where the weighting explicitly back-loads, late-period deviations dominate; in places where the weight shape isn't published, you only know it's time-weighted—you'll have to measure the shape yourself.

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